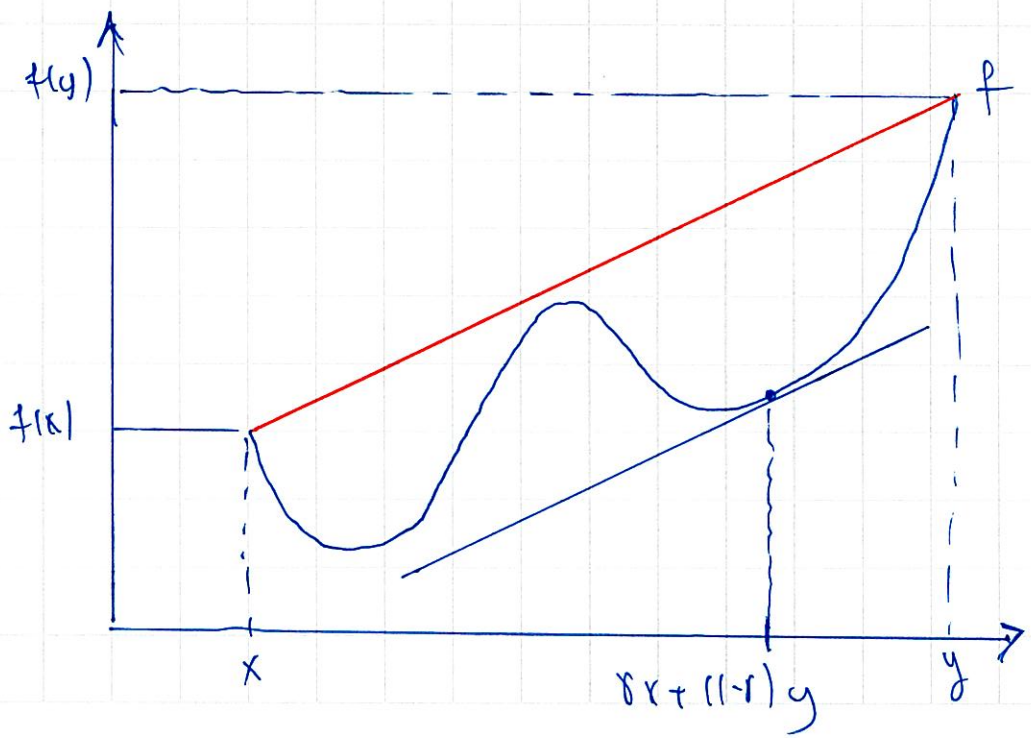
Taylor's Theorem:

- We use the notation $o(t)$ to denote any function $g(t)$ such that $\lim_{t \to 0} \frac{g(t)}{t} = 0$

- Recall $x \in \mathbb{R}^n$, $\|x\| = \sqrt{\sum_{i=1}^{n} x_i^2}$ $\quad x^t . x = (x_1, x_2, \dots, x_n) \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \sum_{i=1}^{n} x_i^2 = \|x\|^2$

- Taylor's Theorem: Let $f: \mathbb{R}^n \to \mathbb{R}$

$$f(y) = f(x) + \nabla^t f(\gamma x + (1-\gamma) y) . (y - x) . \quad \text{for some } \gamma \in (0,1)$$



Implication of Taylor's Theorem:

For any $x, y \in \mathbb{R}^n$, we have

$$f(y) = f(x) + \nabla^t f(x) \cdot (y-x) + o(\|y-x\|) \qquad (\ddagger)$$

Why does Taylor's Theorem imply $\ddagger$?

By Taylor's theorem, for any x, y,

$$f(y) = f(x) + \nabla^t f(\gamma x + (1-\gamma) y) \cdot (y-x) \qquad \text{for some } \gamma \in (0,1)$$

$$= f(x) + \nabla^t f(x)(y-x) + \left(\nabla f^t(\gamma x + (1-\gamma) y) - \nabla f^t(x)\right)(y-x)$$

$$= f(x) + \nabla^t f(x)(y-x) + \|\nabla f^t(\gamma x + (1-\gamma) y) - \nabla f^t(x)\| \cdot \|y-x\| \cos\theta,$$

where θ is the angel between $\nabla f(\gamma x + (1-\gamma) y) - \nabla f(x)$ and $y-x$

We will be done if we can show

$$\|\nabla f(\gamma x + (1-\gamma) y) - \nabla f(x)\| \, \|y-x\| \cdot \cos\theta \quad \text{is} \quad o(\|y-x\|)$$

Check $\lim_{\|y-x\| \to 0} \dfrac{\|\nabla f(\gamma x + (1-\gamma) y) - \nabla f(x)\| \, \cancel{\|y-x\|} \cos\theta}{\cancel{\|y-x\|}} = 0$

Because $\|y-x\| \to 0 \iff y - x \to 0 \iff \gamma x + (1-\gamma) y \to x$

Why is the implication of Taylor's theorem important?

Say we are at point $\hat{x}$ with $\nabla f(\hat{x}) \neq 0$. We have been saying that if we move against $\nabla f(\hat{x})$ for a small step size, the value of the function gets smaller. This result follows from the implication of Taylor's theorem. Why?

implication of Taylor's theorem

$$f(\hat{x} - \alpha \nabla f(\hat{x})) = f(\hat{x}) + \nabla^t f(\hat{x}) \cdot (\cancel{\hat{x}} - \alpha \nabla f(\hat{x}) - \cancel{\hat{x}}) + o(\|\cancel{\hat{x}} - \alpha \nabla f(\hat{x}) - \cancel{\hat{x}}\|)$$

$$= f(\hat{x}) - \alpha \|\nabla f(\hat{x})\|^2 + o(\alpha \|\nabla f(\hat{x})\|)$$

$$= f(\hat{x}) + \alpha \|\nabla f(\hat{x})\| \left(- \|\nabla f(\hat{x})\| + \frac{o(\alpha \|\nabla f(\hat{x})\|)}{\alpha \|\nabla f(\hat{x})\|} \right) \quad (\neq$$

Note: $\lim_{\alpha \to 0} \dfrac{o(\alpha \|\nabla f(\hat{x})\|)}{\alpha \|\nabla f(\hat{x})\|} = 0$, this means that we can pick

$\alpha > 0$ small enough such that $\dfrac{o(\alpha \|\nabla f(\hat{x})\|)}{\alpha \|\nabla f(\hat{x})\|} \leq \dfrac{1}{2} \|\nabla f(\hat{x})\|$

Thus by (†) we get

$$f(\hat{x} - \alpha \nabla f(\hat{x})) \leq f(\hat{x}) + \alpha \|\nabla f(\hat{x})\| \cdot \left(- \|\nabla f(\hat{x})\| + \frac{1}{2} \|\nabla f(\hat{x})\| \right)$$

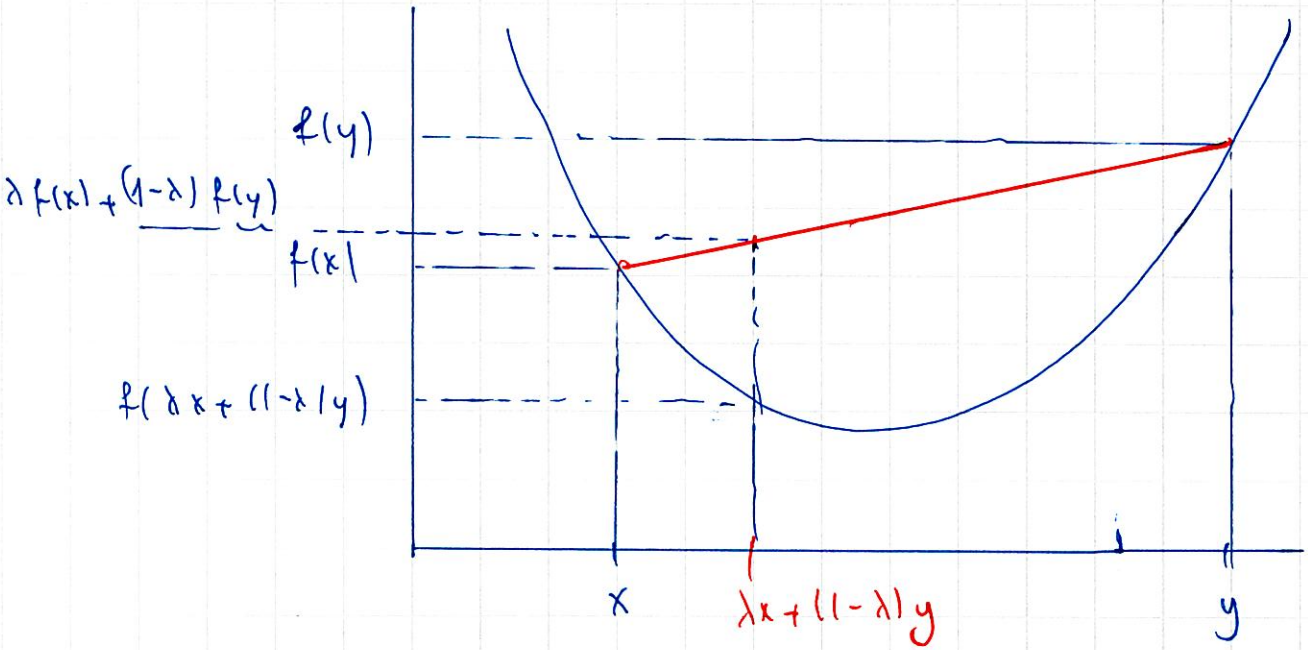
$$= f(\hat{x}) - \frac{1}{2} \alpha \|\nabla f(\hat{x})\|^2 \leq f(\hat{x})$$

Thus, $f(\hat{x} - \alpha \nabla f(\hat{x})) \leq f(\hat{x})$

Convex Functions:

Def: The function $f: \mathbb{R}^n \longrightarrow \mathbb{R}$ is convex if it satisfies:

$$f(\lambda x + (1-\lambda) y) \leq \lambda f(x) + (1-\lambda) f(y) \quad \text{for all } x, y \in \mathbb{R}^n, \ \lambda \in [0,1]$$



Jensen's Inequality

Let $f: \mathbb{R}^n \to \mathbb{R}$ be convex, $x^1, x^2, \dots, x^k \in \mathbb{R}^n$, $\lambda^1, \lambda^2, \dots, \lambda^k \in \mathbb{R}_+$ such that $\sum_{\ell=1}^{k} \lambda^\ell = 1$. Then

$$f\left(\sum_{\ell=1}^{k} \lambda^\ell \cdot x^\ell\right) \leq \sum_{\ell=1}^{k} \lambda^\ell f(x^\ell)$$

Pf: $$f\left(\sum_{\ell=1}^{k} \lambda^\ell x^\ell\right) = f\left(\lambda^1 x^1 + (1-\lambda^1)\sum_{\ell=2}^{k} \frac{\lambda^\ell}{1-\lambda^1} \cdot x^\ell\right)$$

f is convex

$$\leq \lambda^1 f(x^1) + (1-\lambda^1)\, f\left(\sum_{\ell=2}^{k} \frac{\lambda^\ell}{1-\lambda^1} x^\ell\right)$$

$$= \lambda^1 f(x^1) + (1-\lambda^1)\, f\left(\frac{\lambda^2}{1-\lambda^1} x^2 + \sum_{\ell=3}^{k} \frac{\lambda^\ell}{1-\lambda^1} x^\ell\right)$$

$$= \lambda^1 f(x^1) + (1-\lambda^1)\, f\left(\frac{\lambda^2}{1-\lambda^1} x^2 + \left(1-\frac{\lambda^2}{1-\lambda^1}\right) \cdot \sum_{\ell=3}^{k} \frac{\lambda^\ell}{\left(1-\frac{\lambda^2}{1-\lambda^1}\right)(1-\lambda^1)} x^\ell\right)$$

$\to 1-\lambda^1-\lambda^2$

f is convex

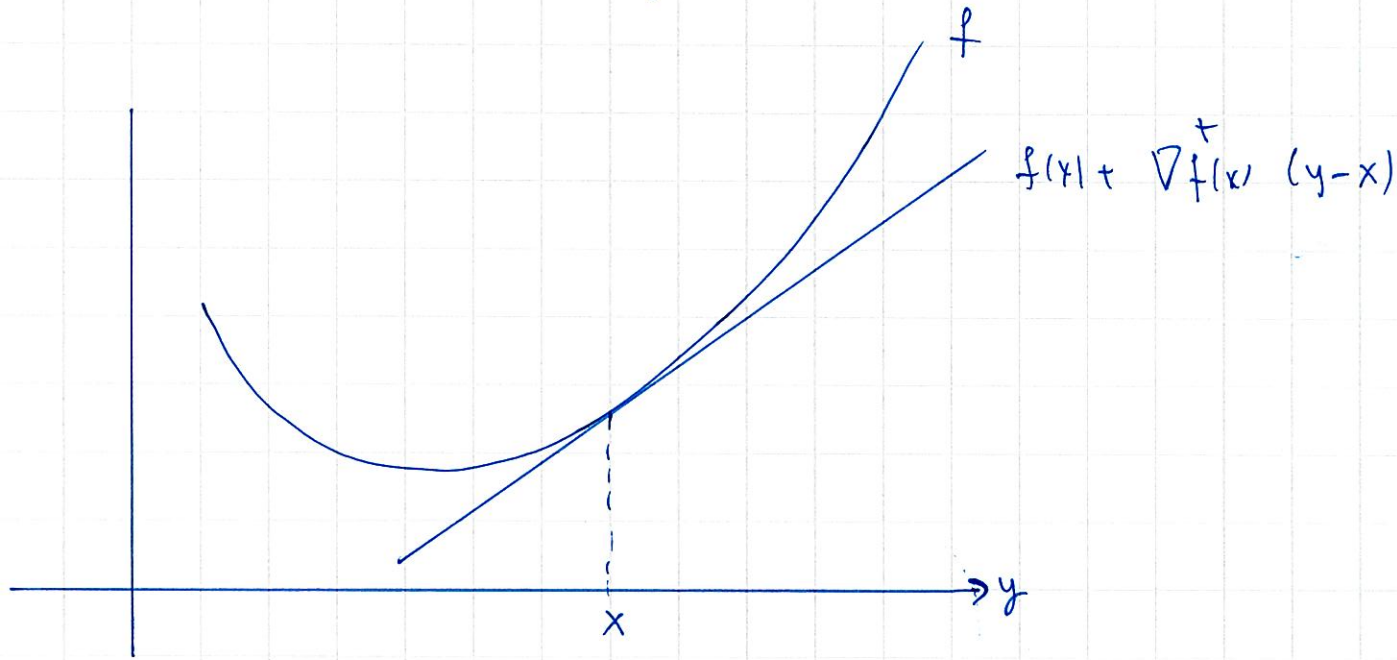
$$\leq \lambda^1 f(x^1) + (1-\lambda^1)\left(\frac{\lambda^2}{1-\lambda^1} f(x^2) + \left(1-\frac{\lambda^2}{1-\lambda^1}\right) f\left(\sum_{\ell=3}^{k} \frac{\lambda^\ell}{1-\lambda^1-\lambda^2} x^\ell\right)\right)$$

$$\frac{1-\lambda^1-\lambda^2}{1-\lambda^1}$$

$$= \lambda^1 f(x^1) + \lambda^2 f(x^2) + (1-\lambda^1-\lambda^2)\, f\left(\sum_{\ell=3}^{k} \frac{\lambda^\ell}{1-\lambda^1-\lambda^2}\, x^\ell \right)$$

Continue similarly

Gradient Based Characterization of Convexity

Theorem: The function $f: \mathbb{R}^n \to \mathbb{R}$ is convex if and only if

$$f(y) \geq f(x) + \nabla^T f(x) \cdot (y-x)$$